

ROS MVP Readiness & Remaining Work

What is genuinely finished, what the first real MVP must contain, and the shortest defensible path from the current repository to it — measured against `R0S_SRS_v1.0.pdf` and the implementation progress audit.

SRS 634 defined requirements Audit baseline 2026-08-16 · 48a16f9 Repo now e5648fb · feat/production-spec

Governance 17 ratified · 1 part · 1 blocked · 1 open

MVP READINESS

NOT READY — but design prerequisites for the critical path are clear

The repository holds a genuinely excellent foundation covering roughly one seventh of the SRS. **Every revenue-generating domain is absent.** No order can be created, no payment taken, no ticket fired, no receipt printed. A restaurant cannot operate on this system today.

The MVP is not blocked by unresolved design. Sales/POS — the critical path — has no open governance decision against it. The three unresolved schema items and the two blocked decisions all sit on Governance approvals, which is *not* on the MVP critical path.

0 Source discipline & discrepancies

Reported, not silently resolved.

DISCREPANCY	DETAIL	MVP IMPACT
The audit baseline is stale	The progress audit is dated 2026-08-16 at commit <code>48a16f92</code> on <code>main</code> . It scores Production Spec as DESIGNED ONLY, 0% and names it “the single largest overcounting risk”. Since then commit <code>896b572</code> implemented Production Spec: migration <code>20260817120000</code> , 5 tables live, 5 Prisma models, 9 routes, a service layer, 1 unit spec and 2 E2E specs. The audit’s headline number and its critical path both predate this.	CORRECTED BELOW
Four dangling requirement IDs	<code>FR-INT-020</code> , <code>FR-PLT-041</code> , <code>FR-RPT-055</code> , <code>FR-SEC-018</code> are referenced in the SRS but never defined. Excluded from the 634 inventory.	DEFERRABLE — resolve in the SRS
Test-count basis differs	The audit reports 384 passing tests (136 unit, 248 E2E) from an executed run. This report did not re-run the suite (no execution permitted), so that figure is carried as the audit’s, not re-verified. File-level counts today: 25 unit specs, 18 E2E specs.	NONE
Source file count	Audit: 221 TypeScript files. This report counts 171 excluding <code>src/generated/</code> . Different counting basis, not a regression.	NONE

1 Current state

BASELINE AS AUDITED (2026-08-16, 48A16F9)

13.3% WEIGHTED PROGRESS	11.6% UNWEIGHTED	50 COMPLETE	46 PARTIAL
12 DESIGNED ONLY	23 BLOCKED	501 NOT IMPLEMENTED	2 OUT OF SCOPE

REVISED SNAPSHOT INCLUDING THE PRODUCTION SPEC DELTA

Applying the audit’s own method to the delta: **FR-MNU-040, 041, 042, 043, 044, 045, 047, 049 and BR-MNU-001** move to COMPLETE; **BR-MNU-002** becomes PARTIAL (versions are retained, but there are no orders to reference them). **FR-MNU-046 [M], FR-MNU-048 [S], FR-MNU-050 [C] and BR-MNU-003** remain NOT IMPLEMENTED — cost expansion was explicitly deferred by D-17-05.

≈14.5% WEIGHTED (ESTIMATED)	≈59 COMPLETE	≈47 PARTIAL	≈3 DESIGNED ONLY
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23

BLOCKED

≈498

NOT IMPLEMENTED

This revised figure is an inference, not a re-audit. It applies the audit's stated weighting to a delta I verified in the repository. It is not a fresh pass over all 634 requirements, and it should not be quoted as one. The audit's 13.3% remains the only fully-derived figure.

VOCABULARY — USED STRICTLY THROUGHOUT

TERM	MEANS
IMPLEMENTED	Executable code exists and runs.
DESIGNED	A design gate or ADR exists. Contributes 0% to progress.
RATIFIED	A governance decision is closed. Says nothing about code.
VERIFIED	Automated tests, or live database probes with positive controls, prove the behaviour.
PRODUCTION-READY	Implemented + verified + the mandatory security and operational controls for that surface are in place.

DOMAINS WITH EXECUTABLE IMPLEMENTATION — 7 OF 29

DOMAIN	WHAT ACTUALLY WORKS	VERIFIED BY	MOST IMPORTANT GAP
Identity & Auth	Email/password (Argon2), 15-min access tokens, 30-day refresh with rotation and reuse detection revoking the whole family, password reset	auth, refresh, password E2E	No PIN auth (FR-SEC-021/022) — the primary POS method. No MFA (FR-SEC-023/024)
Tenancy	RLS ENABLE + FORCE on 62 tables, 213 policies, fail-closed predicate, non-bypassing runtime role	3 dedicated RLS E2E suites, live ros_app probes with positive controls	Tenant lifecycle absent — no self-signup (FR-PLT-020), export (021), termination (023)
RBAC	Roles, permission catalogue, role-permission and membership-role assignment, guard-enforced on all 123 routes	rbac.e2e-spec.ts	Branch scope not enforced — membership_roles.branch_id exists but is never read (FR-SEC-002/003/004)
Audit	SHA-256 per-tenant hash chain, append-only by privilege, advisory-lock serialised	audit.e2e-spec.ts, hash + verify unit specs	3 of 10 FR-AUD complete. No verification job (005), no query surface (007/008), no retention (009)
Organisation	Tenants, brands, branches, warehouses, central kitchens, stations, tables, operating hours, print routing, org.locations registry	organisation.e2e-spec.ts	Branch groups and org.settings deferred → FR-PLT-025_028 settings resolver unimplemented
Catalogue	Menus, categories, items, variants, modifier groups, price lists, availability rules, 86-ing, branch menu resolution	catalogue.e2e-spec.ts, menu-resolution.spec.ts	Price resolution not implemented — valid_from, valid_to, recurrence_rule, priority stored but never evaluated (FR-MNU-021/022/023)
Inventory	Stock items, batches, levels, append-only partitioned movements, transfers with counterpart pairing, counts, waste, FIFO/FEFO, weighted-average/standard/FIFO valuation	inventory.e2e-spec.ts, inventory-rls.e2e-spec.ts	13 FR-INV BLOCKED on Procurement/Sales/Governance; 5 need a scheduler
Production / Recipes	Recipes by scope, versions with single-published invariant, lines with sub-recipe references, cycle detection with full path, substitute groups, publish transition	production.e2e-spec.ts, production-rls.e2e-spec.ts, recipe-graph.spec.ts	No costing — FR-MNU-046 [M] and BR-MNU-003 deferred by D-17-05
NEW SINCE AUDIT			

DOMAINS WITH ZERO IMPLEMENTATION

Sales/POS (76) · Offline & Sync (36) · Procurement (33) · Costing & Analytics (31) · Workforce (29) · Finance & Treasury (27) · CRM & Loyalty (26) · Kitchen Ops/KDS (23) · Reporting (18) · DevOps & Release (15) · Integrations (6) · Quality Assurance (5) · plus most of Localisation and nearly all NFRs.

2 If we stopped development today

CAPABILITY	STATUS	EVIDENCE	MOST IMPORTANT LIMITATION
Authentication / session lifecycle	COMPLETE	Argon2, JWT rotation, reuse detection, family revocation; 4 E2E suites	PIN and MFA missing — POS cannot authenticate an operator
Tenant isolation	COMPLETE	62 tables RLS-forced, 213 policies, live non-bypassing role probes with positive controls	None material. This is the strongest part of the system
RBAC	PARTIAL	All routes guarded; permission catalogue enforced	Tenant-scope only. Branch scope unenforced
Organisation hierarchy	COMPLETE	Full hierarchy + locations registry; ADR 0008, 17 decisions	Settings resolver absent
Catalogue	PARTIAL	13 tables, 40 routes, branch menu resolution tested	Price resolution absent — an order could not be priced
Recipes / Production	PARTIAL	5 tables, 9 routes, 3 spec files	No cost computation; no consumption path (needs orders)
Inventory	PARTIAL	30 tables, append-only ledger, valuation, counts, waste	Depletion never fires — nothing consumes stock
Audit integrity	PARTIAL	Hash chain verified live; append-only by privilege	Unqueryable, unverified by any job, unretained
API surface	PARTIAL	123 routes, all guarded; validation and 404-on-cross-tenant tested	No stable error codes, no idempotency, no pagination cursors, no machine clients
Database / security foundations	COMPLETE	Composite tenant-safe FKs, CHECK constraints, forward-only migrations, zero drift	8 of 14 required schemas do not exist
Any revenue operation	ABSENT	—	No order, payment, ticket, or receipt exists in any form

The honest summary. What exists is a well-engineered, well-verified platform substrate. It is not a restaurant system. A restaurant could configure its organisation, menu and recipes, and could not then sell anything.

3 Domain gap matrix

MVP decisions are derived from SRS dependency, not intuition. Reasons cite requirement IDs.

Domain	Reqs	Cmp	Part	DSG	BLK	NONE	Progress	MVP Decision	Critical Dependencies	Cplx	Reason
Sales / POS	80	0	0	0	0	80	0%	MUST HAVE	Catalogue pricing, PIN auth, Org	XL	FR-P05-001/002 order creation and numbering are the atomic unit of restaurant operation; \$2.5.3 makes the order the event that drives inventory, cost, cash, tax and audit. Nothing downstream exists without it.
Security & Authz	45	4	10	0	0	31	20%	MUST HAVE	—	L	FR-SEC-021/022 PIN is the primary POS authentication method; FR-SEC-002/003/004 branch scope is required the moment a second branch transacts.
Catalogue / Menu	42	6→14	9	12→3	0	13	26%→45%	MUST HAVE	—	M	FR-MNU-021/022/023 price resolution is unimplemented; an order line cannot be priced without it.
Inventory	53	24	9	0	13	7	55%	SHOULD HAVE	Sales, Recipes	M	The ledger exists; depletion on sale (\$2.5.3 step 13) is what makes it operational. 13 requirements are blocked on domains that do not exist.
Platform & Tenancy	22	7	3	0	0	12	45%	SHOULD HAVE	—	M	Isolation is complete. FR-PLT-025-028 settings resolver is needed for per-branch POS configuration.
Audit	10	3	2	0	0	5	40%	SHOULD HAVE	Scheduler	S	FR-AUD-001 is [M] for every state change — orders and payments must emit entries. FR-AUD-005 verification job needs a scheduler.
API Platform	14	2	5	0	0	7	36%	MUST HAVE	—	M	FR-API-020-023 idempotency is mandatory on financially significant endpoints; FR-P05-065 and NFR-REL-011 make it non-optional for payments.
Kitchen Ops (KDS)	23	0	0	0	0	23	0%	SHOULD HAVE	Sales	L	A dine-in order must reach the kitchen. Minimum ticket lifecycle only; the full KDS feature set is deferrable.
Finance & Treasury	32	0	0	0	0	32	0%	SHOULD HAVE	Sales, Payments	L	FR-FIN-001-010 cash session open/close and variance are required to reconcile a shift that took cash. Only the cash-drawer subset is MVP.
Localisation	20	0	1	0	0	19	2%	SHOULD HAVE	Sales	M	FR-LOC-020-022 country pack governs tax and receipt legality; FR-P05-100 requires country-pack receipt elements. One pack only for MVP.
Reporting	18	0	0	0	0	18	0%	SHOULD HAVE	Sales, Finance	M	A branch cannot be run blind. Minimum: daily sales summary and shift close. Full analytics deferred.
Costing & Analytics	31	0	0	0	0	31	0%	CAN DEFER	Sales, Recipes	L	Theoretical-vs-actual is the product differentiator (\$2.5.3) but not required to operate. Recipe cost (FR-MNU-046) is its prerequisite.
Procurement	33	0	0	0	0	33	0%	CAN DEFER	Inventory	L	Stock can be established by opening balances and counts. Purchasing is not required to sell.
Offline & Sync	36	0	0	0	0	36	0%	CAN DEFER	POS, Catalogue, Inventory	XL	Genuinely required for production restaurant use (FR-OFF-001), but an internal MVP can run online-only . The single largest post-MVP workflow.
Workforce	29	0	0	0	0	29	0%	CAN DEFER	—	L	Staff can be modelled as users with roles. Scheduling, attendance and payroll are not required to sell.
CRM & Loyalty	26	0	0	0	0	26	0%	CAN DEFER	Sales	L	Only 9 of 26 are [M]. Walk-in service needs no customer record.
Branch Ops & Central Kitchen	30	2	3	0	10	15	12%	BLOCKED	Production, Sales	L	10 blocked. Central-kitchen distribution is a multi-branch concern, not a single-branch MVP one.
Governance approvals	—	0	0	~14	2 dec.	—	0%	CAN DEFER	Consuming domain	M	Extensively ratified, zero code. D-14 A-1 removed the HTTP surface and no consuming domain exists , so it delivers nothing until POS needs a discount approval.
DevOps & Release	15	0	0	0	0	15	0%	SHOULD HAVE	—	M	FR-SEC-049 dependency scanning and FR-PLT-014 both fail with no CI. Required before production, not before functional work.
NFR — all groups	52	0	4	0	0	48	~2%	SHOULD HAVE	Observability baseline	M	48 of 52 unimplemented and, more importantly, unmeasurable . No NFR can be claimed until a measurement baseline exists.
Integrations / QA / Portability	17	0	0	0	0	17	0%	CAN DEFER	—	M	No SRS dependency from the core sell-a-meal workflow.

4 The first ROS MVP, defined

The smallest coherent product that lets **one branch of one tenant take an order, get paid, feed the kitchen, deplete stock, and close the day** — with tenant isolation and audit intact. Every capability traces to SRS IDs.

#	Capability	Required SRS IDs	Status Now	What is Missing	Hard Blocker
1	Establish a tenant	FR-PLT-001/002/003 , FR-SEC-001/010/011	COMPLETE	Nothing for an internal MVP. Self-signup FR-PLT-020 is post-MVP	No
2	Establish organisation / branch	FR-BRN-001/002 , FR-PLT-001/002	COMPLETE	Nothing	No
3	Configure catalogue & pricing	FR-MNU-001-030 , esp. 021/022/023	PARTIAL	Price resolution engine — valid_from , valid_to , recurrence_rule , priority are stored but never evaluated	YES
4	Configure recipes	FR-MNU-040-045 , BR-MNU-001/002	COMPLETE	Costing (FR-MNU-046 , BR-MNU-003) — not needed to sell	No
5	Operator authentication at the terminal	FR-SEC-021/022 , FR-SEC-028	ABSENT	PIN authentication entirely . Terminal pairing/activation deferred	YES
6	Order entry & lifecycle	FR-P05-001/002/005/010 , FR-P05-020-040 , BR-P05-*	ABSENT	The entire order aggregate — schema, state machine, numbering, lines, modifiers	YES
7	Pricing, tax & totals	FR-FIN-030-032 , BR-FIN-001/003 , BR-CORE-001/002 , FR-LOC-020-022	ABSENT	Tax computation, rounding policy, Money value object (BR-CORE-001/002 unimplemented in application code), one country pack	YES
8	Payment & tender	FR-P05-060/065 , FR-API-020-023 , NFR-REL-011	ABSENT	Tender types (cash + card-external minimum), idempotency infrastructure	YES
9	Receipt	FR-P05-100 , FR-LOC-022	ABSENT	Receipt document generation with country-pack elements	YES
10	Kitchen ticket	FR-KDS-001-010	ABSENT	Minimum ticket lifecycle: fire, display, bump. Routing tables already exist in org	DINE-IN ONLY
11	Inventory depletion on sale	FR-INV-030 , BR-INV-001 , \$2.5.3 step 13	HALF	Ledger and recipes exist; the consumption path from a completed order does not	YES
12	Cash session & day close	FR-FIN-001-010 , esp. FR-FIN-006	ABSENT	Session open/close, declaration, variance. FR-FIN-006 variance approval needs Governance or a documented MVP carve-out	YES
13	Multi-tenant safety	FR-PLT-003/010/011/012 , FR-SEC-045/047	COMPLETE	Must be extended to every new table — not re-built	No
14	Audit of new operations	FR-AUD-001/002/003/006	PARTIAL	Writer exists; orders, payments and cash events must emit entries	YES, AT BUILD TIME
15	Minimum operational visibility	FR-RPT-001-010 subset	ABSENT	Daily sales summary, shift/session report, stock-on-hand view	YES

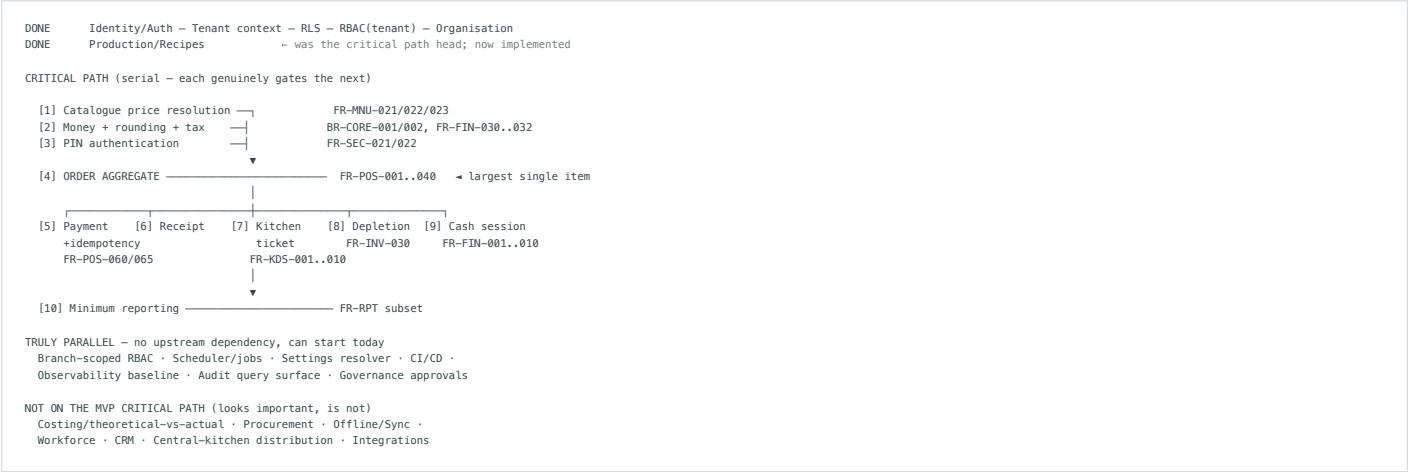
What is deliberately excluded from MVP: offline operation, table floor plans, delivery and aggregator channels, loyalty, procurement, workforce scheduling, theoretical-vs-actual costing, central-kitchen distribution, multi-branch consolidation, integrations. Each is justified in Part 10.

5 MVP blockers, ranked

PRI	WORKSTREAM	REQUIREMENT IDS	WHY IT BLOCKS	UPSTREAM	DOWNSTREAM AFFECTED	D / I / V
P0	Order aggregate & state machine	FR-POS-001/002/005 , FR-POS-020_040 , BR-POS-*	Nothing can be sold. \$2.5.3 makes the completed order the event that drives every other subsystem	Catalogue pricing	Payments, KDS, Inventory, Finance, Costing, Reporting, Offline	none / none / none
P0	Catalogue price resolution	FR-MNU-021/022/023	An order line cannot be priced. Columns exist and are never evaluated	—	Sales, Finance, Reporting	none / partial / none
P0	PIN authentication	FR-SEC-021/022	The primary POS authentication method. An operator cannot identify themselves at a terminal	—	Sales, Finance, SoD	none / none / none
P0	Money value object & rounding	BR-CORE-001/002 , BR-FIN-001/003	Columns are BIGINT minor units but no Money type exists in application code . Currency mismatch and allocation are unenforced	—	All financial computation	none / none / none
P0	Tax computation & one country pack	FR-FIN-030_032 , FR-LOC-020_022	A receipt is not legally issuable without country-pack tax elements	Money	Sales, Finance, Receipts	none / none / none
P0	Payment & idempotency	FR-POS-060/065 , FR-API-020_023 , NFR-REL-011	FR-POS-065 names double-charging the most damaging POS defect class. Mandatory on financially significant endpoints	Order	Finance, Reporting	none / none / none
P0	Inventory depletion on sale	FR-INV-030 , BR-INV-001	Without it the ledger never moves and inventory is decorative	Order, Recipes	Costing, Reporting	none / half / none
P1	Cash session & variance	FR-FIN-001_010	A shift that took cash cannot be closed or reconciled	Payments	Reporting	none / none / none
P1	Receipt generation	FR-POS-100 , FR-POS-104/106	Customers must receive a compliant document	Tax, country pack	—	none / none / none
P1	Kitchen ticket lifecycle	FR-KDS-001_010	A dine-in order must reach the kitchen. Print routing already modelled in org	Order	—	none / none / none
P1	Branch-scoped RBAC	FR-SEC-002/003/004	The column exists and is never read. The moment two branches transact, staff see each other's data	—	All operational domains	none / column only / none
P1	Minimum reporting	FR-RPT-001_010 subset	A branch cannot be operated blind	Sales, Finance	—	none / none / none
P1	Audit emission for new domains	FR-AUD-001/006	[M] for every state change; discounts, voids, refunds and cash variance are named explicitly	Sales	—	n/a / writer exists / none
P2	Scheduler / jobs	FR-AUD-005 , FR-DR-002 , 5 FR-INV	Converts 5 inventory requirements and enables partition creation — a live operational risk	—	Inventory, Audit, DR	none / none / none
P2	Settings resolver	FR-PLT-025_028	Per-branch POS configuration; MVP can hardcode defaults	—	POS, Finance	designed / none / none
P2	CI/CD & dependency scanning	FR-SEC-049 , FR-PLT-014 , FR-OPS-*	No .github exists. Required before production, not before functional work	—	Release	none / none / none
P2	Observability baseline	NFR-005-* , all NFR-PERF-*	No NFR can be <i>claimed</i> without measurement	—	All NFRs	none / correlation id only / none

6 Critical path

The audit's chain was **Production Spec** → **Sales/POS** → **Payments / Kitchen / Costing / Finance / Reporting / Offline**. That was correct when written. **It is now out of date in one respect and imprecise in another.**



7 Implementation phases

Derived from SRS dependency, not assumed. Phase 1 differs materially from the suggested outline because Production/Recipes is already done.

PHASE	GOAL	SRS REQUIREMENTS	PREREQUISITES	DELIVERABLES	VERIFICATION	EXIT CRITERIA
P0	Clear foundations that gate everything financial	BR-CORE-001/002, BR-FIN-001/003, FR-MNU-021/022/023, FR-SEC-021/022	None — all startable today	Money: value object with currency guard and allocate; rounding policy; price-resolution engine; PIN auth with lockout	Unit specs for allocation exactness; price-resolution matrix; PIN lockout E2E	A menu item resolves to a price for a branch at a time; a PIN authenticates an operator
P1	Order aggregate	FR-P05-001_040, BR-P05-*, FR-AUD-001	P0	sales: schema with RLS + composite FKs; order/line/modifier model; branch-day order numbering; state machine; audit emission	RLS E2E mirroring production-rls; state-machine unit specs; numbering uniqueness under concurrency	An order can be opened, itemised, priced and closed within one tenant and branch
P2	Money in, legally	FR-P05-060/065, FR-API-020_023, FR-FIN-030_032, FR-LOC-020_022, FR-P05-100	P1	Tender handling (cash + external card); idempotency key store with replay and fingerprint conflict; tax engine; one country pack; receipt document	Idempotent-replay E2E; tax conformance against the pack; receipt element assertions	A paid order produces a compliant receipt and cannot be double-charged
P3	Operational consequences	FR-INV-030, BR-INV-001, FR-KDS-001_010, FR-FIN-001_010	P2	Recipe-expansion depletion on order completion; kitchen ticket fire/display/bump; cash session open/close/declare/variance	Depletion E2E proving ledger movement equals recipe expansion; session variance E2E	Selling a dish moves stock, fires a ticket, and the shift reconciles
P4	See the branch	FR-RPT-001_010 subset, FR-SEC-002/003/004	P3	Daily sales summary; session report; stock-on-hand; branch-scope enforcement in the guard	Branch-scope isolation E2E with positive controls	A manager can read yesterday's trading, and cannot read another branch's
P5	Production hardening	FR-SEC-023/024, FR-SEC-041/042, FR-SEC-049/050, FR-AUD-005/008, NFR-0B5-*, FR-DR-*	Parallel from P1	MFA for privileged roles; encryption at rest/KMS; secrets injection; CI with dependency scanning; scheduler; audit verification job and query surface; observability baseline; backup/restore	Restore drill; SLO measurement; scan gate	Gate between INTERNAL MVP and PRODUCTION MVP

8 Parallel work

CANDIDATE	CLASSIFICATION	SRS-DEPENDENCY REASONING
Governance approvals	POST-MVP	Extensively ratified, zero code. D-14 A-1 removed the HTTP surface and no consuming domain exists , so it cannot be exercised. It becomes required when POS implements FR-P05-047 discount approval or FR-FIN-006 cash variance — both P1/P2 of the MVP. Reclassified from the audit's "parallel, unblocked" to post-MVP , because building it now delivers nothing callable.
Scheduler / jobs	BEFORE PRODUCTION	Converts 5 FR-INV from PARTIAL, enables FR-AUD-005 and FR-DR-002 automatic partition creation. Partition creation is a live operational risk — the ledger has no DEFAULT partition and inserts past the last boundary fail hard. Cheapest high-leverage item in the repository.
Settings resolver	BEFORE PRODUCTION	FR-PLT-025_028. POS needs per-branch configuration, but an MVP can carry defaults in code. Deferring is safe; skipping is not.
Branch-scoped RBAC	MVP REQUIRED NOW	FR-SEC-002/003/004. membership_roles.branch_id exists and is never read. Start it in parallel immediately — retrofitting scope enforcement after Sales/POS exists is materially harder than building it alongside.
CI/CD	BEFORE PRODUCTION	FR-SEC-049 dependency scanning and FR-PLT-014 both fail with no pipeline. No functional dependency, so it need not gate P0-P4 — but it must precede any production claim.
Observability	BEFORE PRODUCTION	NFR-0B5-*. No NFR-PERF requirement can be asserted without it — 16 performance requirements are currently unmeasurable, not merely unmet.
Procurement	POST-MVP	33 requirements, unblocks only 2 FR-INV. Stock can be established by opening balances and counts, both implemented. Genuinely parallel but low MVP value.
Audit query surface	BEFORE PRODUCTION	FR-AUD-007/008 [M]. Recorded in the governance register as knowingly unmet. An auditor cannot read the log today.

9 Security & control gates

The SRS supports distinguishing an **internal MVP** (a controlled pilot on non-production data) from a **production MVP** (real money, real customers). Nothing mandatory is waived — some controls are sequenced after functional work.

CONTROL	REQUIREMENT	STATE	GATE	NOTE
Tenant isolation	FR-PLT-003/010/011/012	COMPLETE	Done	Must be extended to every new schema, never re-derived
Server-side authorization	FR-SEC-045	COMPLETE	Done	All 123 routes guarded
Input validation	FR-SEC-047	COMPLETE	Done	Whitelist + forbid-non-whitelisted, tested
Authentication & session	FR-SEC-020, FR-API-010/013	COMPLETE	Done	Rotation with reuse detection
Audit integrity	FR-AUD-002/003/004	COMPLETE	Done	Hash chain verified live
PIN authentication	FR-SEC-021/022	MISSING	BLOCKS MVP	Primary POS auth method
Branch scope	FR-SEC-002/003/004	MISSING	BLOCKS MVP	Multi-branch data separation
Idempotency	FR-API-020_023, NFR-REL-011	MISSING	BLOCKS MVP	Mandatory on financial endpoints; double-charge is the named worst defect
Audit emission for new domains	FR-AUD-001/006	PARTIAL	BLOCKS MVP	Discounts, voids, refunds, cash variance named explicitly
Segregation of duties	FR-SEC-015/016/017	MISSING	INTERNAL → PROD	FR-SEC-016 requires blocking. Needed once approvals exist
Approval workflow	FR-SEC-030_035	DESIGNED	INTERNAL → PROD	Needed for FR-P05-047 / FR-FIN-006. Carve-out must be explicit if deferred
MFA	FR-SEC-023/024	MISSING	BEFORE PRODUCTION	Mandatory for security.user.manage, settings.tenant.manage, api.key.manage
Rate limiting (per-tenant)	FR-PLT-015, FR-SEC-046	PARTIAL	BEFORE PRODUCTION	IP-based on auth only; no per-tenant limits, no progressive lockout
Encryption at rest / KMS	FR-SEC-041/042/043	MISSING	BEFORE PRODUCTION	No envelope encryption, no per-tenant keys
Secrets management	FR-SEC-050	MISSING	BEFORE PRODUCTION	Configuration via .env only
Dependency scanning / CI	FR-SEC-049, FR-PLT-014	MISSING	BEFORE PRODUCTION	No .github exists
Observability	NFR-0B5-*	PARTIAL	BEFORE PRODUCTION	Correlation id only; no structured logging, tracing, RED metrics or alerting

CONTROL	REQUIREMENT	STATE	GATE	NOTE
Performance SLOs	NFR-PERF-« (16)	UNMEASURED	BEFORE PRODUCTION	Unmeasurable, not merely unmet
Backup / restore	FR-DR-020/021	MISSING	BEFORE PRODUCTION	No per-tenant restore capability
Data retention & privacy	FR-SEC-060_064 , FR-AUD-009	MISSING	POST-MVP	Required before real customer data
IP allow-listing / SIEM	FR-SEC-052/053	MISSING	POST-MVP	Enterprise-tier controls

Non-negotiable. Four controls must ship *with* the functional MVP, not after: **PIN authentication, branch scope, idempotency on payments, and audit emission for orders, payments and cash.** Each is [M] and each is materially harder to retrofit than to build alongside.

10 What to defer

CAPABILITY	SRS IDS	WHY DEFERRABLE	RISK OF DEFERRAL	LATER PHASE
Offline & Sync	FR-OFF-001_047	An internal MVP runs online-only. No other MVP capability depends on it	High. 34 [M] requirements; retrofitting offline into an online-only design is expensive. Design for it early even if built late	Post-MVP 1
Costing / theoretical-vs-actual	FR-CST-« , FR-HNU-046 , BR-HNU-003	The differentiator, but not required to operate. Already deferred by D-17-05	Medium — it is the product's stated value proposition	Post-MVP 1
Procurement	FR-PRC-«	Stock established via opening balances and counts, both implemented	Low	Post-MVP 2
Workforce	FR-HRM-«	Staff modelled as users with roles	Low	Post-MVP 2
CRM & Loyalty	FR-CRM-«	Only 9 of 26 are [M]; walk-in service needs no customer record	Low	Post-MVP 2
Table management / floor plan	FR-P05-000_090	Mostly [S]. Orders can carry a table reference without a graphical plan	Low	Post-MVP 1
Delivery, drive-through, aggregators	FR-P05-001 subset, FR-INT-«	MVP covers dine-in and takeaway only	Medium — aggregator revenue matters commercially	Post-MVP 1
Central-kitchen distribution	FR-BRN-« subset	Multi-branch concern; single-branch MVP does not need it	Low	Post-MVP 2
Full reporting & analytics	FR-RPT-011_055	MVP needs daily sales, session close and stock-on-hand only	Low	Post-MVP 1
Additional country packs	FR-LOC-023_030	One pack proves the architecture	Low	Post-MVP 2

11 False-completion risks

Assessed on executable evidence, not document existence. The audit identified nine; this reassessment updates two and adds one.

CLAIM	REALITY	EVIDENCE
"Production Spec is ratified and authorized"	NOW PARTLY BUILT	Updated. The audit scored this DESIGNED/0% and was right at the time. It is now implemented — but still not complete : FR-HNU-046 [M] cost recomputation and BR-HNU-003 remain unbuilt, deferred by D-17-05. Do not report Production as done.
"Phase 15 Organisation complete"	MOSTLY	Complete for entities defined, but D-02 branch-scoped RBAC, D-10 branch groups and D-11 org.settings were deferred, leaving FR-SEC-002/003/004 , FR-BRN-005 and FR-PLT-025_028 unimplemented.
"Phase 16 Catalogue complete"	PARTIAL	13 tables and 40 routes exist, but price resolution is not implemented — valid_from , valid_to , recurrence_rule , priority are stored and returned, never evaluated. This is an MVP P0 blocker hiding inside a "complete" phase.
"Inventory complete, 248 E2E passing"	PARTIAL	22 of 50 FR-INV/ complete; 13 blocked; 5 need a scheduler; 7 dropped by explicit decision.
"RLS is fully enforced"	TRUE NOW	True today — but it was false when inventory was first reported complete . 14 ledger partitions had no RLS of their own, leaking every tenant's ledger to a direct-partition query. Caught by a live probe, not the test suite. Direct evidence that green tests did not equal a secure boundary.
"Audit trail satisfies FR-AUD"	3 OF 10	Hash chain and append-only storage are excellent; no verification job (005), no query/search/export surface (007/008), no retention (009), no impersonation capture (010).
"Rate limiting implemented"	PARTIAL	IP-based throttling on authentication endpoints only. FR-PLT-015 per-tenant limits and FR-SEC-046 progressive lockdown both absent.
"Multi-tenant SaaS platform"	0 OF 4	Tenant isolation is excellent; tenant lifecycle is absent — no self-signup (020), no export (021), no downgrade (022), no termination (023).
"Passing build / green suite"	NOT COVERAGE	Passing tests prove the implemented slice works. They cannot prove absent domains exist. Around 498 requirements have no code to test.
"Governance is nearly done"	NEW0% CODE	Added by this report. Governance has 17 ratified decisions, a ratified parent-linkage decision, and a large register — and zero executable code : no approval tables, no service, no route, no test. governance contains only audit_entries . Ratification volume is the newest false-completion risk in the project.

12 Remaining MVP effort

Relative engineering size only. The repository contains no velocity evidence, so no calendar estimate is defensible.

WORKSTREAM	SIZE	BASIS
Money value object + rounding	S	Well-specified in §7.2 with exact semantics; pure logic, no schema
Catalogue price resolution	M	Columns exist; resolution and recurrence semantics are non-trivial
PIN authentication	S	Salted hash + branch uniqueness + lockout; auth patterns already established
Branch-scoped RBAC	M	Column exists; guard, token claims and every query path must honour scope
Order aggregate	XL	New schema, state machine, numbering, lines, modifiers, discounts, voids. 80 requirements
Tax + one country pack	L	Multi-component tax, rounding points, pack loading and signing
Payment + idempotency	L	Tender handling plus a general idempotency layer used system-wide
Receipt generation	M	Document assembly against pack-mandated elements
Kitchen ticket lifecycle	M	Minimum fire/display/bump; routing already modelled

WORKSTREAM	SIZE	BASIS
Inventory depletion on sale	M	Recipe expansion exists; the consumption path and its transaction boundary do not
Cash session & variance	M	Session lifecycle, declaration, variance, tolerance
Minimum reporting	S	Three read models over existing and new tables
Scheduler / jobs	S	Small capability, disproportionate leverage
Production-gate security bundle	L	MFA, KMS, secrets, CI, observability, backup/restore

SEQUENCES

Optimistic — price resolution, Money and PIN run in parallel; the order aggregate proceeds without rework; tax needs only one pack; depletion reuses the existing ledger cleanly.

Realistic — the order aggregate churns once as pricing, tax and discount interactions surface; idempotency forces a cross-cutting refactor of existing write paths; branch scope forces revisiting query paths already written; the first country pack takes longer than expected because it is also the pack *architecture*.

UNCERTAINTY FACTORS

- No client exists.** Every requirement here is backend-only; a POS UI is an entirely separate and unestimated effort.
- Idempotency is cross-cutting** and touches routes already built.
- Country-pack architecture** is designed but unproven; the first pack validates or invalidates it.
- Offline design pressure** — building online-only risks an expensive retrofit later. Design decisions taken now should not preclude it.

13 MVP requirement ledger

MVP-relevant requirements only. Traceable to the SRS.

PRI	REQUIREMENT	SUBSTANCE	DOMAIN	STATUS	DEPENDENCY	IMPLEMENTATION NEEDED	VERIFICATION	BLK
P0	BR-CORE-001/002	Currency guard; exact allocation	Core	NONE	—	Money value object	Allocation exactness specs	Y
P0	BR-FIN-001/003	Round once; distribute discount exactly	Finance	NONE	Money	Rounding policy	Unit specs	Y
P0	FR-MNU-021/022/023	Price list resolution, validity, priority	Catalogue	STORED ONLY	—	Resolution engine	Resolution matrix specs	Y
P0	FR-SEC-021/022	PIN auth, branch-unique, lockout	Security	NONE	—	PIN credential + lockout	Lockout E2E	Y
P0	FR-POS-001/002	Order types; branch-day order number	Sales	NONE	P0	sales schema + numbering	Uniqueness under concurrency	Y
P0	FR-POS-005/010	Concurrent open orders; item selection	Sales	NONE	Order	Order/line model	E2E	Y
P0	FR-POS-020-040	Lines, modifiers, discounts, voids	Sales	NONE	Order	Line lifecycle + state machine	State-machine specs	Y
P0	FR-PLT-003/010/011/012	Tenant isolation on new tables	Platform	PATTERN READY	—	Extend RLS to sales	RLS E2E with positive controls	Y
P0	FR-AUD-001/006	Audit every state change	Audit	WRITER ONLY	Order	Emission in sales paths	Entry-per-operation E2E	Y
P0	FR-FIN-030-032	Multi-component tax	Finance	NONE	Money	Tax engine	Conformance vs pack	Y
P0	FR-L0C-020/021/022	Country pack; signed; rejected if invalid	Localisation	NONE	Tax	Pack loader + one pack	Signature rejection test	Y
P0	FR-POS-060	Tender types	Sales	NONE	Order	Cash + external card	E2E	Y
P0	FR-POS-065	Payments idempotent	Sales	NONE	Payment	Idempotency layer	Replay E2E	Y
P0	FR-API-020-023	Idempotency-Key, replay, 409 on mismatch	API	NONE	—	Key store + middleware	Fingerprint-conflict E2E	Y
P0	NFR-REL-011	At-most-once financial effect	NFR	NONE	Idempotency	—	Concurrency test	Y
P1	FR-POS-100	Compliant customer receipt	Sales	NONE	Tax, pack	Receipt document	Element assertions	Y
P0	FR-INV-030 · BR-INV-001	Movements on sale; append-only	Inventory	LEDGER ONLY	Order, Recipes	Depletion via recipe expansion	Ledger-equals-expansion E2E	Y
P1	FR-KDS-001-010	Ticket fire, display, bump	Kitchen	NONE	Order	Ticket lifecycle	E2E	Dine-in
P1	FR-FIN-001-010	Cash session, declaration, variance	Finance	NONE	Payment	Session lifecycle	Variance E2E	Y
P1	FR-SEC-002/003/004	Branch scope; no cross-scope leakage	Security	COLUMN ONLY	—	Guard + token + query scope	Branch isolation E2E	Y
P1	FR-RPT-001-010 subset	Daily sales, session, stock on hand	Reporting	NONE	Sales, Finance	Three read models	Figures reconcile to ledger	Y
P2	FR-AUD-005	Scheduled chain verification + alert	Audit	NONE	Scheduler	Job + alerting	Break-detection test	N
P2	FR-AUD-007/008	Audit access audited; searchable, exportable	Audit	NONE	Read surface	Query + export	Permission-gated E2E	N
P2	FR-SEC-023/024	MFA for privileged roles	Security	NONE	—	TOTP enrolment	E2E	N
P2	FR-SEC-041/042/043	Envelope encryption, per-tenant keys	Security	NONE	KMS	Field encryption	Round-trip test	N
P2	FR-SEC-049 · FR-PLT-014	Dependency scanning; CI gates	DevOps	NONE	—	Pipeline	Failing-scan gate	N
P2	FR-PLT-015 · FR-SEC-046	Per-tenant limits; progressive lockout	Platform	PARTIAL	—	Extend throttling	Limit E2E	N

PRI	REQUIREMENT	SUBSTANCE	DOMAIN	STATUS	DEPENDENCY	IMPLEMENTATION NEEDED	VERIFICATION	BLK
P2	FR-DR-002 · FR-DR-020/021	Partition automation; per-tenant restore	DR	NONE	Scheduler	Jobs + restore path	Restore drill	N
P2	NFR-OBS-* · NFR-PERF-*	Structured logs, tracing, RED, SLOs	NFR	UNMEASURED	—	Observability baseline	SLO measurement	N

14 Executive summary

1 · WHAT WE HAVE NOW

A rigorously built platform substrate covering roughly one seventh of the SRS: authentication with token rotation, tenant isolation enforced by 213 RLS policies across 62 tables and verified live against a non-bypassing role, tenant-scoped RBAC on all 123 routes, a tamper-evident audit chain, the full organisation hierarchy, a catalogue, an append-only inventory ledger with valuation, and — new since the audit — recipes with versioning and cycle detection. **Seven of twenty-nine domains have executable code.**

2 · WHAT IS MISSING FOR MVP

Everything that makes money. No order, no price resolution, no PIN login, no `Money` type, no tax, no payment, no receipt, no kitchen ticket, no stock depletion, no cash session, no reporting. **Nothing in the system can complete a sale.**

3 · THE TRUE MVP SCOPE

One tenant, one brand, one branch. Dine-in and takeaway. Staff log in by PIN, build an order from a priced menu, take cash or an external card payment, print a compliant receipt, fire a ticket to the kitchen, deplete stock by recipe, and close the cash session — with tenant isolation, branch scope and audit intact, and a daily sales summary to read the next morning. **Online-only. Single country pack.**

4 · THE #1 CRITICAL PATH

Price resolution + Money + PIN → Order aggregate → Payment + idempotency → depletion, tickets, cash close → minimum reporting. The order aggregate is the pivot: roughly 230 requirements sit downstream of it.

5 · WHAT CAN RUN IN PARALLEL

Start now: branch-scoped RBAC (hardest to retrofit), scheduler/jobs (cheapest leverage, and it defuses a live partition risk), observability baseline, CI/CD. **Anytime:** audit query surface, settings resolver.

6 · WHAT WE SHOULD NOT BUILD YET

Governance approvals — 17 ratified decisions and zero callable code; it delivers nothing until POS needs discount approval. **Offline/Sync, Costing, Procurement, Workforce, CRM, central-kitchen distribution, integrations, floor plans, additional country packs.**

7 · MVP EXIT CRITERIA

- 1. A staff member authenticates by PIN at a terminal bound to a branch.
- 2. An order is created, priced from a resolved price list, taxed by country pack, and closed.
- 3. Payment is taken and **cannot be double-charged under retry.**
- 4. A compliant receipt is produced.
- 5. A dine-in order fires a kitchen ticket that can be bumped.
- 6. Completing the order **moves the stock ledger** by recipe expansion.
- 7. The cash session closes with a computed variance.
- 8. A manager reads a daily sales summary and **cannot read another branch's.**
- 9. Every one of the above emits an audit entry, and the chain still verifies.
- 10. Cross-tenant access returns 404 everywhere, proven by live probes with positive controls.

8 · THE FIRST TEN WORK ITEMS, IN ORDER

- 1. `Money` value object — `BR-CORE-001/002`, `BR-FIN-001/003`
- 2. Catalogue price resolution — `FR-MNU-021/022/023`
- 3. PIN authentication — `FR-SEC-021/022`
- 4. Branch-scoped RBAC — `FR-SEC-002/003/004` (*parallel; retrofit cost rises daily*)
- 5. Scheduler/jobs — `FR-DR-002` first, to defuse the partition-boundary risk
- 6. `saLes` schema + RLS + composite FKs — `FR-PLT-003/010/011/012`
- 7. Order aggregate & state machine — `FR-P05-001_040`
- 8. Idempotency layer — `FR-API-020_023`
- 9. Tax engine + first country pack — `FR-FIN-030_032`, `FR-LOC-020_022`
- 10. Payment & tender — `FR-P05-060/065`

NOT READY

Not ready as a product — no MVP capability can be exercised end to end. **But the design prerequisites for the MVP critical path are sufficiently resolved:** Sales/POS, price resolution, PIN, Money, tax and payments carry *no* open governance decision. The three unresolved schema items and both blocked decisions sit on Governance approvals, which this report places off the MVP path.

Implementation of the MVP critical path can begin immediately. The blocker is build effort, not unresolved design.

? Unknown / unresolved

Matters the SRS does not decide, or that this report cannot settle from the sources. None is resolved here.

- **Four dangling requirement IDs** — `FR-INT-020`, `FR-PLT-041`, `FR-RPT-055`, `FR-SEC-018` are referenced but never defined in the SRS.
- **Internal vs production MVP** — the SRS does not formally define this distinction. Part 9 uses it as a sequencing device; **it is inference, not an SRS concept**, and adopting it is a business decision.
- **Which country pack ships first** — the SRS defines the architecture, not the launch jurisdiction.
- **Whether MVP may defer the approval workflow** — `FR-SEC-030` is [M], and `FR-POS-047` / `FR-FIN-006` both reference approval. Deferring requires an **explicit, recorded carve-out**; this report does not grant one.
- **Governance schema items** — `required_permission` FK posture, `value` persistence type and nullability, and the `approval_requests` DELETE posture remain unresolved in the register. **Off the MVP path, so not MVP blockers.**
- **D-16** remains OPEN and **D-12** remains BLOCKED. **P-1 stands: `approval_decisions` references `approval_requests` directly.** Governance tally unchanged at **17 ratified · 1 in part · 1 blocked · 1 open**.
- **Client/UI effort is entirely unestimated.** Every figure here is backend-only.
- **The 384-test baseline was not re-verified** — no suite was executed for this report.

Sources: `R05_SRS_v1.0.pdf` (634 defined requirements) · `R05_SRS_IMPLEMENTATION_PROGRESS_AUDIT.pdf` (2026-08-16, `48a16f92`) · repository and live database state at `e5648fb` · `GOVERNANCE_DECISION_REGISTER.md`. Analysis only — no implementation, no schema, migration, RLS, database, or governance change was made.